Transfer Learning and Model Fine-Tuning Analysis: A ResNet-18 Case Study

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Abstract

In this study, an image classification model was constructed using the CIFAR-10 dataset and a pre-trained ResNet-18 architecture. ResNet-18 was selected to achieve a balance between computational complexity and model efficiency appropriate for low-resolution input images. Experiments evaluated four distinct hyperparameter configurations, systematically varying batch size, learning rate, and epoch duration to observe independent effects on convergence and accuracy. The baseline model achieved an overall test accuracy of 95%. Higher learning rates and batch sizes did not improve accuracy and increased overall training time, while extending training to 20 epochs demonstrated convergence without overfitting at 15 epochs. Class-level error analysis revealed consistent misclassification between visually similar categories, specifically cats and dogs. Potential mitigation strategies, including class-targeted data augmentation and weighted loss functions, are discussed.

Index Terms Transfer learning, image classification, ResNet-18, CIFAR-10, hyperparameter tuning, error analysis.

I. Introduction

For this assignment, an image classification model was built using the CIFAR-10 dataset and the ResNet-18 pretrained model architecture. ResNet-18 was selected for this project on account of providing an effective balance between model complexity and computational efficiency appropriate for the task. ResNet-18 was specifically chosen over larger architectures such as ResNet-50 or EfficientNet because those deeper models appeared excessive given that CIFAR-10 images are of low resolution. To prevent overfitting, 18 layers of depth were determined to be sufficient for the dataset. Furthermore, ResNet-18 offered lightweight computational requirements capable of training efficiently on cloud GPU hardware without sacrificing classification accuracy.

II. Hyperparameter Configuration Comparison

To study the effects of different training parameters on model performance, four distinct configurations were evaluated, focusing on variations in batch size, learning rate, and number of training epochs. Configuration 1 served as the baseline model. For Configurations 2, 3, and 4, a single hyperparameter was isolated and modified to evaluate its independent impact on the network.

TABLE I: HYPERPARAMETER CONFIGURATIONS

Configuration	Batch Size	Learning Rate	Epochs	Overall Test Accuracy
1	16	0.001	15	95%
2	16	0.01	15	94%
3	24	0.001	15	95%
4	16	0.001	20	95%

A. Impact of Learning Rate (Config 1 vs. Config 2)

Increasing the learning rate from 0.001 to 0.01 resulted in a performance degradation, reducing overall test accuracy from 95% down to 94%. Despite taking larger optimization steps, Configuration 2 required approximately 3.5 additional minutes of training time compared to Configuration 1. The learning rate of 0.01 proved too large to permit smooth convergence to the global minimum, indicating it was less suitable for this architecture.

B. Impact of Batch Size (Config 1 vs. Config 3)

Increasing the batch size from 16 to 24 provided no improvement in overall test accuracy, remaining constant at 95%. While larger batch sizes can provide more accurate gradient estimates per optimization step, the baseline batch size of 16 was sufficient to maintain training stability. Furthermore, increasing the batch size to 24 increased total training duration, reducing overall training efficiency without yielding accuracy gains.

C. Impact of Epochs (Config 1 vs. Config 4)

Extending total training duration from 15 to 20 epochs resulted in plateaued accuracy at 95%. This indicates that the ResNet-18 model successfully converged around epoch 15. Additional training epochs consumed computational resources and

extended training duration without discovering new feature patterns. Notably, extended training did not induce overfitting, though it provided no practical benefit for model performance.

III. Error Analysis

Despite achieving a high overall test accuracy of 95%, class-specific evaluations revealed distinct performance disparities across categories. Across all tested configurations, misclassifications concentrated primarily within two specific classes. The cat category consistently recorded the lowest classification accuracy, ranging between 89.1% and 89.4%. The dog category exhibited the second-lowest accuracy, ranging from 92.7% to 94.1%. Conversely, non-animal categories achieved high accuracy scores; for instance, the automobile class ranged between 97.9% and 98.3%, while the frog class reached 98.2% to 98.4%.

The primary driver for confusion between cats and dogs stems from shared visual features, including quadrupedal postures, fur textures, similar body dimensions, and comparable ear and tail structures. Background environment similarities and pose variations likely contributed to feature confusion. Furthermore, the low spatial resolution of CIFAR-10 images obscures fine-grained visual details such as facial structure and fur patterns, causing the network to struggle when differentiating between visually similar categories.

IV. Recommendations for Improvement

To improve overall classification accuracy beyond 95%, targeted strategies should focus on mitigating misclassifications within the cat and dog classes. First, class-specific data augmentation techniques (such as random cropping, rotation, and color jittering applied specifically to cat and dog samples) could assist the model in learning invariant feature representations rather than relying on background or pose cues. Second, modifying the loss function using class-weighted Cross-Entropy loss would penalize cat and dog misclassifications more heavily, encouraging the optimization process to prioritize feature resolution for vulnerable categories.

Config 1: Batch 16, LR 0.001, Epochs = 15

Accuracy of the network on test images: 95%

TABLE II: CLASS-SPECIFIC ACCURACY FOR CONFIGURATION 1 BASELINE MODEL

Class	Accuracy
plane	96.1%
car	97.9%
bird	94.4%
cat	89.4%
deer	96.5%
dog	93.8%
frog	98.4%
horse	97.2%
ship	97.8%
truck	97.2%

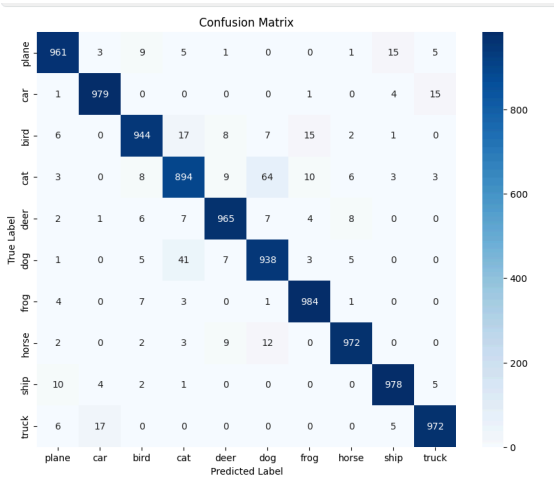


Fig. 1. Confusion matrix for Configuration 1 baseline model on CIFAR-10 test dataset.



Fig. 2. Sample predictions and ground truth labels for Configuration 1.

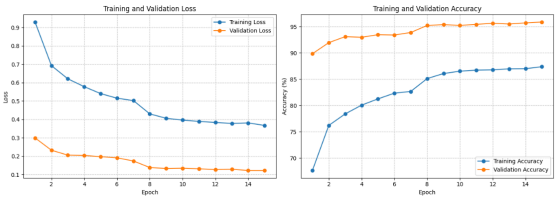


Fig. 3. Training and validation loss/accuracy curves across 15 epochs for Configuration 1.

Config 2: Batch Size = 16 | LR = 0.01 | Epochs = 15

Accuracy of the network on test images: 94 %

TABLE III: CLASS-SPECIFIC ACCURACY FOR CONFIGURATION 2 MODEL

Class	Accuracy
plane	94.5 %
car	96.6 %
bird	93.0 %
cat	89.3 %
deer	94.4 %
dog	92.1 %
frog	96.2 %
horse	96.5 %
ship	96.9 %
truck	96.3 %

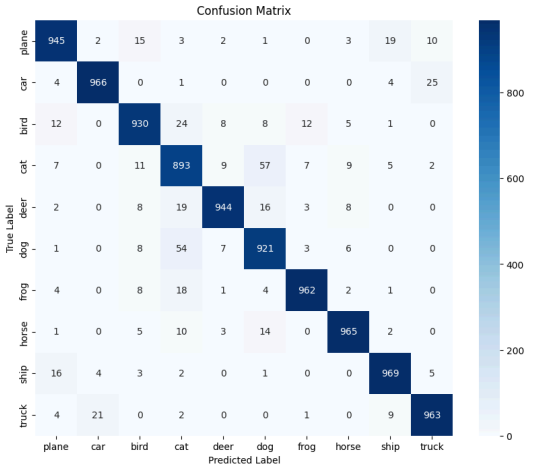


Fig. 4. Confusion matrix for Configuration 2 (learning rate = 0.01).



Fig. 5. Sample predictions and ground truth labels for Configuration 2.

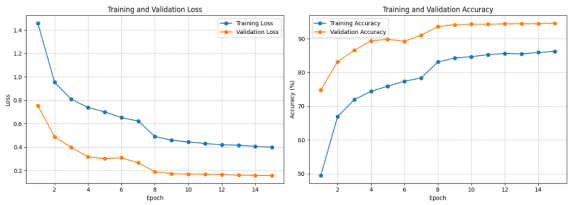


Fig. 6. Training and validation loss/accuracy curves across 15 epochs for Configuration 2.

Config 3: Batch Size = 24 | LR = 0.001 | Epochs = 15

Accuracy of the network on test images: 95 %

TABLE IV: CLASS-SPECIFIC ACCURACY FOR CONFIGURATION 3 MODEL

Class	Accuracy
plane	96.0 %
car	98.0 %
bird	94.5 %
cat	89.1 %
deer	96.5 %
dog	92.7 %
frog	98.2 %
horse	96.7 %
ship	96.9 %
truck	96.6 %

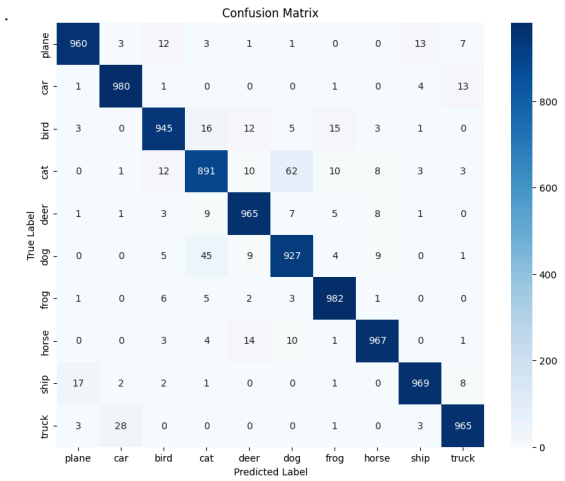


Fig. 7. Confusion matrix for Configuration 3 (batch size = 24).



Fig. 8. Sample predictions and ground truth labels for Configuration 3.

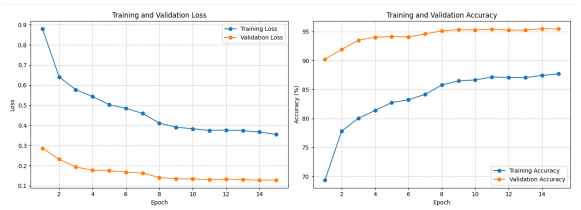


Fig. 9. Training and validation loss/accuracy curves across 15 epochs for Configuration 3.

Config 4: Batch number 16, Learning Rate: 0.001, Epochs = 20

Accuracy of the network on test images: 95 %

TABLE V: CLASS-SPECIFIC ACCURACY FOR CONFIGURATION 4 MODEL

Class	Accuracy
plane	96.3 %
car	98.3 %
bird	94.9 %
cat	89.2 %
deer	95.9 %
dog	94.1 %
frog	98.2 %
horse	96.9 %
ship	97.8 %
truck	96.6 %

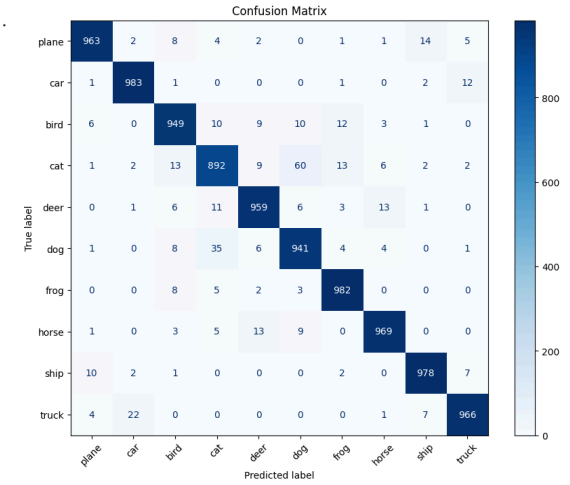


Fig. 10. Confusion matrix for Configuration 4 (epochs = 20).

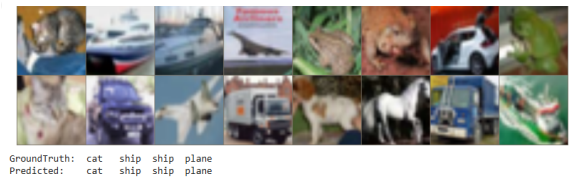


Fig. 11. Sample predictions and ground truth labels for Configuration 4.

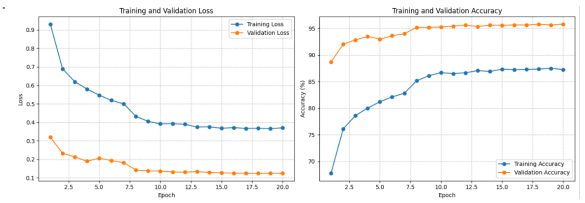


Fig. 12. Training and validation loss/accuracy curves across 20 epochs for Configuration 4.